IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Production all-KCL 200-s switching and SG-reference handback

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Abstract

One six-state EMF6 synchronous generator (SG) and four registered full-state dual-mode IBRs are simulated by the project-owned fixed-step, all-KCL hybrid engine for 200 s. The exact chronology is SG trip at 20 s, load increase at 50 s, bus-9 fault at 85–85.15 s, line 6–13 trip at 110 s, topology restoration and SG-close request at 145 s, and endpoint at 200 s. The operating point uses a 1.00 pu slack reference with a ±5% admissible voltage band. The run converges with maximum accepted-step residual 9.99×10^{-9} and at most one subdivision level. The synchronism guard accepts the SG at 146.825 s; all IBRs then return from GFM to GFL through the separately declared coordinated SG-reference handback. By the 200 s endpoint the SG frequency has settled to 60.000 Hz, though five remote load buses remain slightly below the ±5% voltage band. No Bayesian-optimisation controller is used. A seeded band-limited display ripple is drawn beside, not instead of, each raw trace and affects figures only.

1 Case, decision index, and switching semantics

The system base is 100 MVA and 60 Hz. The SG is at bus 1; IBR1–IBR4 are at buses 2, 3, 6, and 8. The buses carry a parallel `bus_role` label (`REF`/`PV`/`PQ`/`GFM`/`GFL`) that is *presentation-only* and never feeds the power-flow equations (Section 1): the SG bus 1 is REF, buses 2/3/6/8 (the IBRs) are GFL resources on this operating point, and the remaining load buses are PQ. The numerical internal bus type (REF= 1, PV= 2, PQ= 3) in the 12-column `bus_data` is unchanged by this descriptor. The index that produced the results in this report is the seven-term AGSI++ average

$$J_V = \frac{|V - V_{ref}|}{0.10}, \qquad J_f = \frac{|f - 60|}{0.50}, \qquad J_R = \frac{|\mathrm{d}f/\mathrm{d}t|}{1.0}, \tag{1}$$

$$J_P = \frac{|P_{ref} - P|}{0.20}, \qquad J_{SCR} = \max(0, 3/SCR - 1), \qquad J_{lock} = \frac{|v_q|}{0.10}, \tag{2}$$

$$J_{GRA} = 1 - GRA, \qquad \mathrm{AGSI}_i^{++} = \min\left(1, \max\left(0, \frac{1}{7}\sum J_k\right)\right), \tag{3}$$

where $GRA = 1$ when an online SG or committed GFM supplies a reference. Its first four bases are source-guided; the equal 1/7 weighting of the remaining three terms is `PROJECT_DERIVED` and is the part of (3) whose provenance cannot be defended term by term. Section 1.1 states the redesigned index that replaces it; the numbers in this report predate that redesign and were produced by (3). A design-time weight tuner exists in the repository (`scripts/reporting/optimize_agsi_wei`

NUMERICAL_METHOD, random search plus coordinate descent). Its output was *not* used: the reported run uses equal $1/7$ weights.

The generic local state machine uses $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$, $T_{d,on} = 0.10$ s, and $T_{d,off} = 1.00$ s. In this declared chronology, however, the SG-trip transaction is a CASE_DEFINED all-GFM override, and the return at SG close is a coordinated handback. Neither transaction is presented as an index-threshold crossing.

1.1 Redesigned decision index: severity plus hard gates

The switching decision is about restoring a voltage and frequency reference, so the redesign keeps only those two quantities in the continuous objective and demotes everything else to an admissibility test. The severity is

$$S_i = \min\bigl(1, \max\bigl(0,\ w_V J_V + w_f J_f\bigr)\bigr), \qquad J_V = \frac{|V - V_{ref}|}{\Delta V_{base}}, \quad J_f = \frac{|f - f_0|}{\Delta f_{base}}, \tag{4}$$

with $\Delta V_{base} = 0.10$ pu, $\Delta f_{base} = 0.50$ Hz, and $w_V = w_f = 0.5$, $w_V + w_f = 1$. The two bases are SOURCE_DEFINED; the equal split follows the source ratio $0.30 : 0.30$ renormalised, so no free weight remains.

Every other former term becomes a hard not-to-exceed gate returning $L \in \{0, 1\}$, never a weighted cost:

Table 1: Admissibility gates. A mode change requires every gate admissible.

Gate	Test	Meaning	Classification
RoCoF	$\lvert \mathrm{d}f/\mathrm{d}t \rvert \le 1.0\ \mathrm{Hz\,s^{-1}}$	rate admissibility	SOURCE_DEFINED
Power tracking	$\lvert P_{ref} - P \rvert \le 0.20$ pu	dispatch feasibility	SOURCE_DEFINED
Grid strength	$SCR \ge SCR_{crit}$	GFL model validity	SOURCE_DEFINED
PLL lock	$\lvert v_q \rvert \le 0.10$ pu	synchronisation health	PROJECT_DERIVED
Reference	$GRA = 1$	a forming reference exists	PROJECT_DERIVED
Current	$\lvert I \rvert \le I_{max}$	converter limit	SOURCE_DEFINED

The switching law is then

$$\text{GFL} \to \text{GFM}: \quad S_i \ge \Gamma_{on} \wedge \textstyle\bigwedge_k L_k \ \text{ for } T_{d,on}, \qquad \text{GFM} \to \text{GFL}: \quad S_i \le \Gamma_{off} \wedge \textstyle\bigwedge_k L_k \ \text{ for } T_{d,off}, \tag{5}$$

retaining the existing hysteresis and dwell values. Each weight is now either source-defined or fixed by normalisation, and each remaining quantity is a published limit rather than a tuned coefficient. This design is frozen here *before* any re-run; equations (4)–(5) are not used to produce any number in Sections 4–6 of this report.

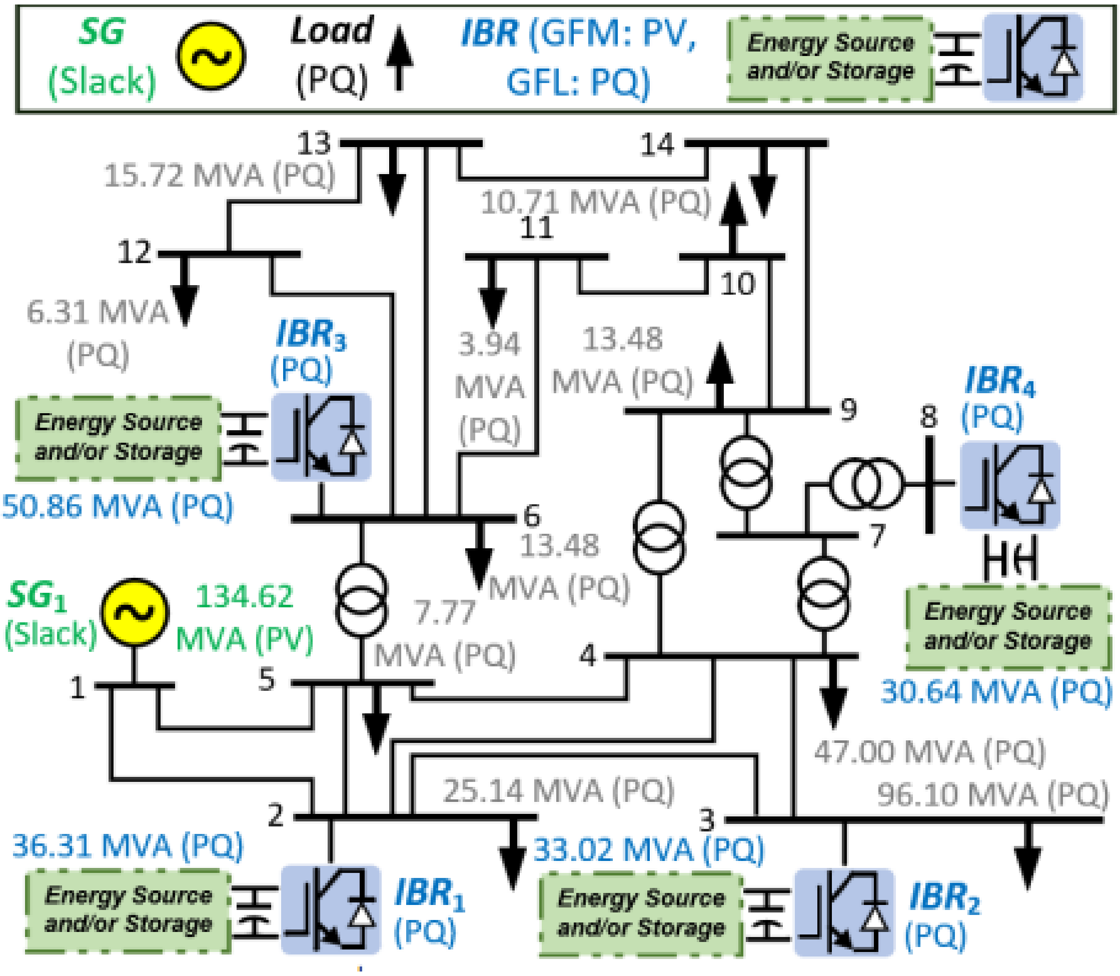

Figure 1: Network and resource locations used by the simulation.

2 Dynamic equations

The operational SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$:

$$\dot{\delta} = \omega_0 \Delta\omega, \qquad 2H\dot{\Delta\omega} = P_m - P_e - D\Delta\omega, \tag{6}$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \qquad T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \tag{7}$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \qquad T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \tag{8}$$

The source singular limit $T'_{q0} = 0$ freezes $E'_d = 0$ exactly. Two prime-mover states remain outside the six-state electromagnetic order:

$$T_{SV}\dot{P}_{SV} = -P_{SV} + P_c - \Delta\omega/R, \qquad T_{CH}\dot{P}_m = -P_m + P_{SV}.$$

They provide offline runback/phase preparation and post-close primary regulation. The numerical mapping is diagnostic and is not a machine-specific governor validation.

2.1 IBR state superset

Each IBR carries a constant-dimension **20-state** switching superset: the 13-state grid-forming branch (REGFM_B1 generator model G2) and the 7-state grid-following branch (WECC REGC_A/REEC_A) both exist at all times,

$$x_{GFM} = [\Delta\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, x_{PLL}, P_{inv,f}, I_{d,f}, Q_{inv,f}, V_{inv,f}, I_{q,f}, \delta_{IT,max}, \delta_{IT,min}]^T, \quad (9)$$

$$x_{GFL} = [V_{t,f}, P_f, I_{q,cmd,f}, P_{ord}, V_{LVPL,f}, I_{p,reg}, I_{q,reg}]^T, \quad (10)$$

so that $x_{IBR} = [x_{GFM}^T \; x_{GFL}^T]^T \in \mathbb{R}^{20}$. All quantities are per-unit on the device base M_{base}; the network interface applies $\kappa = S_{base}/M_{base}$. Powers are positive into the network. Only the branch selected by the present mode q contributes a nonzero derivative; the other branch satisfies $\dot{x} = 0$ exactly.

2.2 Grid-forming branch: 13 states, term by term

The angle reference is the sum of a phase-locked component and a virtual-machine component, so the GFM branch *does* own a PLL pair. With $V = V_x + \mathrm{j}V_y$ the terminal voltage and $v_q = -V_x \sin\delta_{PLL} + V_y \cos\delta_{PLL}$ its quadrature error,

$$\dot{x}_{PLL} = v_q, \qquad \dot{\delta}_{PLL} = \omega_0 \left(k_{p,PLL} v_q + k_{i,PLL} x_{PLL}\right), \quad (11)$$

both frozen to zero while $|V| < V_{PLL,frz}$ so that a collapsed terminal cannot drive the tracker. The virtual synchronous machine supplies the frequency deviation, with a washout state providing transient-only damping:

$$2H\dot{\Delta\omega}_m = P_{ref,inv} - P_{inv,f} - \left(1/m_p + D_1\right)\Delta\omega_m - D_2\left(\Delta\omega_m - x_w\right), \quad (12)$$

$$\dot{x}_w = \omega_D\left(\Delta\omega_m - x_w\right). \quad (13)$$

Here $1/m_p$ is the droop gain, D_1 the steady damping, and D_2 acts only on the high-pass component $\Delta\omega_m - x_w$. The internal angle is stored *relative* to the PLL, which is what keeps the two angle sources consistent:

$$\dot{\delta}_{IT} = \omega_0\Delta\omega_m - \dot{\delta}_{PLL}, \qquad \delta_{VSM} = \delta_{PLL} + \delta_{IT}, \quad (14)$$

with $\dot{\delta}_{IT}$ held at zero whenever δ_{IT} sits on the active bound of $[\delta_{IT,min}, \delta_{IT,max}]$ and the derivative would push it further out. The voltage magnitude comes from a PI regulator with reactive droop and anti-windup,

$$\dot{x}_E = V_{ref} - V_{inv,f}, \qquad E_{VSM}^{raw} = V_{ref} - m_q Q_{inv,f} + k_{pv}\left(V_{ref} - V_{inv,f}\right) + k_{iv} x_E, \quad (15)$$

where $E_{VSM} = \mathrm{sat}\left(E_{VSM}^{raw}, E_{min}, E_{max}\right)$ and $\dot{x}_E$ is set to zero when the limiter is saturated in the direction of further integration. The current-angle bounds are themselves states, integrating the mismatch between the steady-state d-axis current limit and the measured d-axis current, so that the angle authority contracts when the device is overloaded:

$$\dot{\delta}_{IT,max} = k_I\left(I_{d,max}^{SS} - I_{d,f}\right) \in [0, \delta_{max}], \quad \dot{\delta}_{IT,min} = k_I\left(-K_e I_{d,max}^{SS} - I_{d,f}\right) \in [-\delta_{max}, 0], \quad (16)$$

each frozen on its own bound by the same conditional-hold rule, with $\delta_{max} = \arcsin\left(X_L I_{max}^{SS}\right)$ and $\dot{\delta}_{IT,min} = 0$ when the energy-storage flag is inactive. The five remaining states are first-order measurement filters,

$$T_{pf}\dot{P}_{inv,f} = \kappa P - P_{inv,f}, \qquad T_{If}\dot{I}_{d,f} = \kappa I_d - I_{d,f}, \qquad T_{Qf}\dot{Q}_{inv,f} = \kappa Q - Q_{inv,f}, \quad (17)$$

$$T_{Vf}\dot{V}_{inv,f} = |V| - V_{inv,f}, \qquad T_{If}\dot{I}_{q,f} = \kappa I_q - I_{q,f}, \quad (18)$$

where $P + jQ = V\overline{I_{out}}$ and (I_d, I_q) are the injected current projected on the PLL frame. The network injection is voltage-behind-impedance with circular saturation:

$$I_{out} = \begin{cases} I_{unc}, & |I_{unc}| < I^F_{max}/\kappa, \\ \dfrac{I^F_{max}}{\kappa}\dfrac{I_{unc}}{|I_{unc}|}, & \text{otherwise,} \end{cases} \qquad I_{unc} = \frac{E_{VSM}e^{j\delta_{VSM}} - V}{\kappa\,(R_e + jX_L)}. \tag{19}$$

2.3 Grid-following branch: 7 states, term by term

The GFL branch is the registered WECC pair and is a *current-source* model: it owns no PLL state because it aligns algebraically to the measured terminal-voltage angle. Its injection is

$$I = \frac{\left(I_{p,reg}\,\gamma(|V|) - jI_{q,reg}\right)e^{j\angle V}}{\kappa}, \tag{20}$$

with γ the REGC_A low-voltage-point gain and a high-voltage reactive clamp applied above V_{olim}. The states are three measurement filters, the power order, and the two regulated currents:

$$T_{rv}\dot{V}_{t,f} = |V| - V_{t,f}, \qquad T_p\dot{P}_f = \kappa P - P_f, \qquad T_{fltr}\dot{V}_{LVPL,f} = |V| - V_{LVPL,f}, \tag{21}$$

$$T_{pord}\dot{P}_{ord} = \text{sat}\left(\kappa P_{ref}, P_{min}, P_{max}\right) - P_{ord}, \quad T_{iq}\dot{I}_{q,cmd,f} = I_{q,cmd} - I_{q,cmd,f}, \tag{22}$$

where $\dot{P}_{ord}$ is additionally rate-limited to $[\dot{P}_{min}, \dot{P}_{max}]$. The REEC_A command pair divides the order by the filtered voltage and adds the voltage-dip reactive injection,

$$I_{p,0} = \frac{P_{ord}}{\max\left(V_{t,f}, 0.01\right)}, \qquad I_{q,0} = \frac{\kappa Q_{ref}}{\max\left(V_{t,f}, 0.01\right)} + I_{q,inj}, \tag{23}$$

$$I_{q,inj} = \text{sat}\left(K_{qv}\,\text{db}\left(V_{ref,0} - V_{t,f}\right), I_{ql1}, I_{qh1}\right) \qquad \text{active only if } V_{t,f} < V_{dip} \text{ or } V_{t,f} > V_{up}, \tag{24}$$

followed by the active-priority allocation $I_p = \text{sat}(I_{p,0}, 0, I_{max})$ and $|I_q| \le \min\left(I_{max}, \sqrt{I^2_{max} - I^2_p}\right)$. The regulated currents then track their commands through the converter lag with the sourced ramp limits,

$$T_g\dot{I}_{p,reg} = I^{tgt}_p - I_{p,reg} \quad \left(\dot{I}_{p,reg} \le r_{rpwr}\right), \quad T_g\dot{I}_{q,reg} = I_{q,cmd,f} - I_{q,reg} \quad \left(\dot{I}_{q,reg} \in [I_{qr,min}, I_{qr,max}]\right), \tag{25}$$

where I^{tgt}_p is I_p passed through the LVPL characteristic when that switch is enabled and floored at zero below V_{lvpnt0}. The 0.01 pu division guard is `NUMERICAL_METHOD`; every gain, time constant, and limit above is `SOURCE_DEFINED`.

2.4 Mode dispatch

In GFL mode the active indices are 14:20 and $\dot{x}_{1:13} = 0$; in GFM mode the active indices are 1:13 and $\dot{x}_{14:20} = 0$; in the tripped mode all twenty derivatives are zero and the injection is identically zero. Only the active branch enters the dynamic solve, and the active-index set is queried from the device rather than hard-coded, so the layout above is not duplicated in the integrator. Atomic state mapping is followed by a full-KCL right-limit solve; a failed solve rolls the transaction back.

3 Switching formulation: what is solved at a mode change

This section states the mathematics executed at every switching instant. It is the governing formulation for the SG trip, the GFL↔GFM transfers, and the SG reclose.

3.1 The plant is a DAE, not an ODE

Between events the plant is a semi-explicit index-1 differential–algebraic system, not a plain ordinary differential equation:

$$\dot{x} = f_q(x, y, u), \tag{26}$$
$$0 = g_q(x, y, u) = Y_q V - I_{inj}(x, y). \tag{27}$$

Here x collects every device differential state solved inside the step (SG six electromagnetic states plus $4{\times}20$ IBR states, i.e. 86 states); the two prime-mover states of Section 2 are advanced by a separate analytic exponential update and enter (26) only through the input u, not as Newton unknowns. Accordingly u carries T_m and E_{fd}, y is the network state stored as rectangular bus voltages, and q is the *discrete mode*: the tuple of network topology, per-device mode, and breaker status. Equation (27) is the full all-KCL residual over all $2n_b$ real rows; a single module owns it, while each device contributes only its own slice of f_q and its own I_{inj}.

The discretisation is the implicit trapezoidal rule, so a normal step is not the explicit recursion $x_{k+1} = x_k + h\, f(\cdot)$ but the simultaneous nonlinear system

$$x_{k+1} - x_k - \tfrac{h}{2}\left[f_q(x_k, y_k, u) + f_q(x_{k+1}, y_{k+1}, u)\right] = 0, \tag{28}$$
$$g_q(x_{k+1}, y_{k+1}, u) = 0, \tag{29}$$

solved together for (x_{k+1}, y_{k+1}). The EMF6 route uses a fixed corrector count of three. Both f_q and g_q carry the subscript q: they are the equations *of the present discrete mode*. A switching event changes q, hence changes the equations themselves.

3.2 The transition at a switching instant

Equations (28)–(29) are *not* continued across an event. At a switching instant t_e the trajectory undergoes a hybrid-DAE transition in three ordered stages:

$$q^- \longrightarrow q^+ \qquad \text{discrete jump,} \tag{30}$$
$$x(t_e^+) = R_{q^-\to q^+}\big(x(t_e^-), y(t_e^-)\big) \qquad \text{state reset map,} \tag{31}$$
$$0 = g_{q^+}\big(x(t_e^+), y(t_e^+), u\big) \quad \text{solve for } y(t_e^+) \qquad \text{right-limit algebraic solve.} \tag{32}$$

Stage (30) selects the post-event admittance matrix Y_{q^+} directly from the declared topology set by its identifier, and commits the device-mode and breaker maps as a single transaction: the whole update is validated before any field is written, and a failed validation leaves the state identical to its input. Stage (31) is described below. Stage (32) solves *only* the algebraic block, by damped Newton iteration with a backtracking line search, holding x fixed at $x(t_e^+)$ and accepting the result only if the KCL residual meets the step tolerance; otherwise the whole transaction rolls back. Differential integration then resumes from $\big(x(t_e^+), y(t_e^+)\big)$ at the next step.

Consequently two samples exist at t_e: a left limit and a right limit. The tables and figures in this report store the right limit.

3.3 The reset map is terminal continuity, not state copying

The key structural property is that **the state dimension never changes**. Each IBR is a constant-dimension 20-state superset in which the GFM states 1:13 and the GFL states 14:20 always exist. A mode change re-selects which sub-block is *active*; the inactive block is held exactly, $\dot{x}_{inactive} = 0$, under a frozen inactive-state rule (`PROJECT_DERIVED`: continuity plus algebraic-residual minimisation). The integrator and the Jacobian therefore never resize. This is also why a later SG reclose is meaningful: nothing was structurally deleted at the trip.

The reset map R in (31) transfers *physical terminal conditions*, not state coordinates, and is owned by the device:

$$V_{bus} = y_{2b-1} + \mathrm{j}y_{2b}, \qquad I^- = I_{inj}(x^-, y, u, q^-), \tag{33}$$

$$S^- = V_{bus}\,\overline{I^-}, \qquad P^- = \Re S^-, \quad Q^- = \Im S^-, \tag{34}$$

$$x^+\big|_{target} = \mathcal{I}_{q^+}(V_{bus}, P^-, Q^-), \qquad x^+\big|_{inactive} = x^-\big|_{inactive}, \tag{35}$$

followed by the fail-closed acceptance test

$$\left|I_{inj}(x^+, y, u, q^+) - I^-\right| \le 10^{-10}. \tag{36}$$

$\mathcal{I}_{q^+}$ is the target branch's own equilibrium/transfer initialiser. The invariant carried across the switch is therefore the terminal pair (V, I) — equivalently (P, Q) — and not the state vector. No PLL angle is copied out of the GFL branch, because the GFL branch owns no PLL state; the GFM angle is re-derived from V_{bus} and I^-. The bus index b is resolved from the device's registered bus position, so the map contains no hard-coded bus identity.

3.4 The SG trip and the gated reclose

At the trip the SG states are *not* removed from x. The breaker-open branch replaces the connected-stator algebra with open-circuit physics: $I_d = I_q = 0$, $T_e = 0$, mechanical torque frozen at its pre-trip value, and open-circuit flux decay,

$$2H\dot{\Delta\omega} = T_m^{froz} - D\Delta\omega, \qquad T''_{d0}\dot{E}''_q = E'_q - E''_q, \qquad T''_{q0}\dot{E}''_d = E'_d - E''_d, \tag{37}$$

i.e. the $(X'_d - X''_d)I_d$ and $(X'_q - X''_q)I_q$ terms vanish with the stator current while $\dot{\delta} = \omega_0\Delta\omega$ continues. The rotor coasts, which is what makes a later resynchronisation physically meaningful.

The reclose is *gated*, not scheduled. A guard compares the network bus phasor against the SG open-circuit EMF $\left(E''_q - \mathrm{j}E''_d\right)e^{\mathrm{j}\delta}$ and requires

$$|\Delta V| \le 0.05 \text{ pu}, \qquad |\Delta f| \le 0.001 \text{ pu}, \qquad |\Delta\theta| \le 10^\circ, \tag{38}$$

expressed as the signed margin $\min\left(0.05 - |\Delta V|,\ 0.001 - |\Delta f|,\ 10^\circ - |\Delta\theta|\right) \ge 0$ sustained for a 0.5 s dwell. The requested close time is therefore only the earliest instant at which closure *may* occur: the request at 145.000 s produced an accepted close at 146.825 s.

4 Power-flow operating point

The power-flow is solved by the project Newton–Raphson routine. Q-limit enforcement is switched on, and the generator reactive-capability limits behind it are *source-defined*, not assumed or tuned: they are the IEEE14 generator data $Q_{\max}$ and $Q_{\min}$ in MVAr (`cases.case_matpower6_case14` `mpc.gen` columns 5 and 4), converted to per unit with the single system base $S_{\text{base}} = 100$ MVA,

$$Q_{\text{max,pu}} = \frac{Q_{\text{max,MVAr}}}{100}, \qquad Q_{\text{min,pu}} = \frac{Q_{\text{min,MVAr}}}{100}, \tag{39}$$

with P_{max} taken the same way from `mpc.gen` column 9. Buses carrying no finite generator limit are left unconstrained ($Q_{\text{max}} = +\infty$, $Q_{\text{min}} = -\infty$) and print a dash.

It must be stated plainly that *enabling the mechanism is not the same as constraining this operating point*. The PV→PQ Q-limit switch is defined only for PV buses, and after the inverter buses are mapped to PQ resources this case has no PV bus at all, so the switch resolves in zero rounds and changes nothing. Two consequences follow and are reported rather than masked: the slack bus 1 settles at $Q_g = -0.196$ pu against its registered $Q_{\text{min}} = 0$, and bus 6 at $Q_g = 0.2476$ pu against $Q_{\text{max}} = 0.24$. Neither is a solver error — a REF bus absorbs whatever reactive power closes the network balance and a PQ resource injects its scheduled Q — but neither is a limit the solve enforced. The limit columns therefore report registered source data, not satisfied constraints. The enforcement path itself is exercised by the baseline IEEE14 case, which does carry PV buses.

The admissible voltage band is ±5% about the 1.00 pu slack reference, i.e. $V_{\text{max}} = 1.05$ pu and $V_{\text{min}} = 0.95$ pu on every bus (`CASE_DEFINED` in `case_ieee14bus_eecon49_switch`). This is a reporting and admissibility band, not a PF equation input: the slack is the only voltage-controlled bus in this operating point, so the band does not act as a PV-setpoint clamp. One observation is reported rather than masked: bus 6 solves at 1.0575 pu, 7.5×10^{-3} pu above V_{max}, because its mapped IBR6 reactive dispatch and the fixed 5–6 tap ratio are source/case-defined and were not adjusted to satisfy the band.

The bus roles (REF/PQ/GFL) in the Type column are the `bus_role` labelling descriptor (Section 1); being a label only, they do not alter (39) or any solved quantity.

Table 2: Bus power-flow results (pu)

Bus	Type	$\lvert V\rvert$	θ (deg)	P_g	Q_g	P_L	Q_L	Q_{min}	Q_{max}	P_{max}
1	REF	1.0000	0.0000	1.3319	-0.1957	0.0000	0.0000	0.00	0.10	3.32
2	GFL	0.9914	-3.2585	0.3172	0.1767	0.2170	0.1270	-0.40	0.50	1.40
3	GFL	0.9667	-8.6921	0.2884	0.1607	0.9420	0.1900	0.00	0.40	1.00
4	PQ	0.9832	-6.7064	0.0000	0.0000	0.4780	-0.0390	–	–	–
5	PQ	0.9858	-5.5221	0.0000	0.0000	0.0760	0.0160	–	–	–
6	GFL	1.0575	-6.7136	0.4443	0.2476	0.1120	0.0750	-0.06	0.24	1.00
7	PQ	1.0253	-6.9205	0.0000	0.0000	0.0000	0.0000	–	–	–
8	GFL	1.0494	-4.4090	0.2677	0.1491	0.0000	0.0000	-0.06	0.24	1.00
9	PQ	1.0219	-8.6416	0.0000	0.0000	0.2950	0.1660	–	–	–
10	PQ	1.0202	-8.5886	0.0000	0.0000	0.0900	0.0580	–	–	–
11	PQ	1.0347	-7.7782	0.0000	0.0000	0.0350	0.0180	–	–	–
12	PQ	1.0411	-7.6862	0.0000	0.0000	0.0610	0.0160	–	–	–
13	PQ	1.0342	-7.8345	0.0000	0.0000	0.1350	0.0580	–	–	–
14	PQ	1.0087	-9.3374	0.0000	0.0000	0.1490	0.0500	–	–	–

Table 3: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.910918	-0.153112	0.016392	-0.002303
2	1	5	0.420987	-0.042553	0.009593	-0.008907
3	2	3	0.466697	0.013221	0.010470	0.002115
4	2	4	0.316733	-0.064476	0.006065	-0.014738
5	2	5	0.211276	-0.049810	0.002649	-0.025732
6	3	4	-0.197333	-0.018164	0.002803	-0.005014
7	4	5	-0.448842	0.084724	0.002882	0.009089
8	4	7	0.018419	-0.096315	0.000000	0.001990
9	4	9	0.062955	-0.012297	0.000000	0.002223
10	5	6	0.092297	0.001911	0.000000	0.001920
11	6	11	0.130821	0.059907	0.001758	0.003682
12	6	12	0.086088	0.027130	0.000895	0.001864
13	6	13	0.207667	0.085524	0.002983	0.005875
14	7	8	-0.267650	-0.134127	0.000000	0.015018
15	7	9	0.286069	0.035823	0.000000	0.008698
16	9	10	-0.003258	0.021664	0.000015	0.000039
17	9	14	0.057282	0.023368	0.000466	0.000991
18	10	11	-0.093273	-0.036375	0.000790	0.001849
19	12	13	0.024192	0.009266	0.000137	0.000124
20	13	14	0.093739	0.030791	0.001556	0.003168

5 200-s event and mode evidence

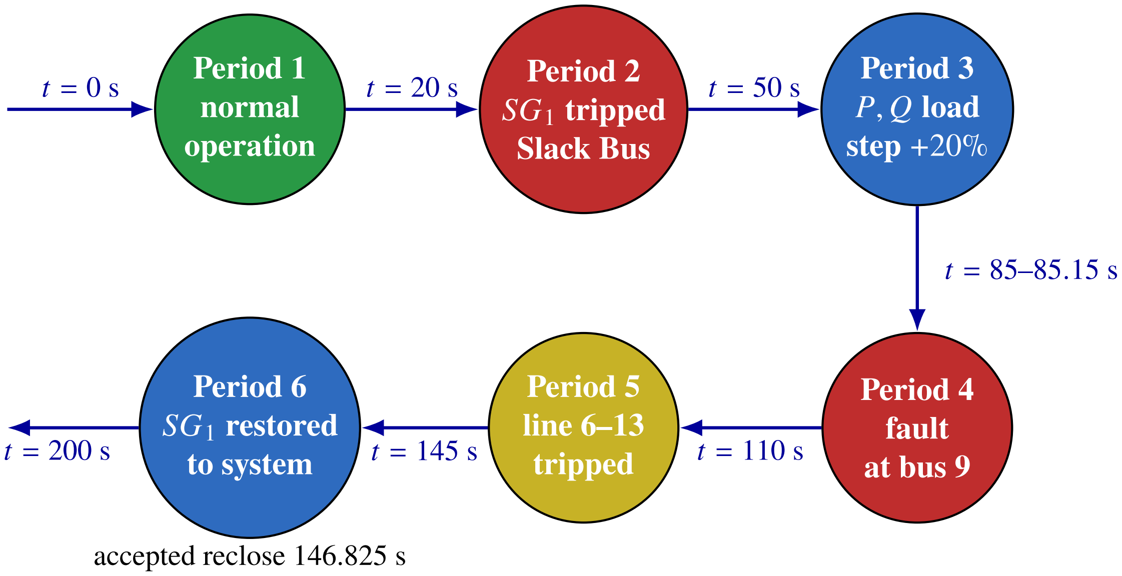

Figure 2: Six-period disturbance sequence of the 200-s simulation from the accepted production event log: normal operation at 0 s, SG trip at 20 s, P, Q load step +20% at 50 s, bus-9 fault 85–85.15 s, line 6–13 trip at 110 s, and SG restoration at 145 s (accepted reclose 146.825 s).

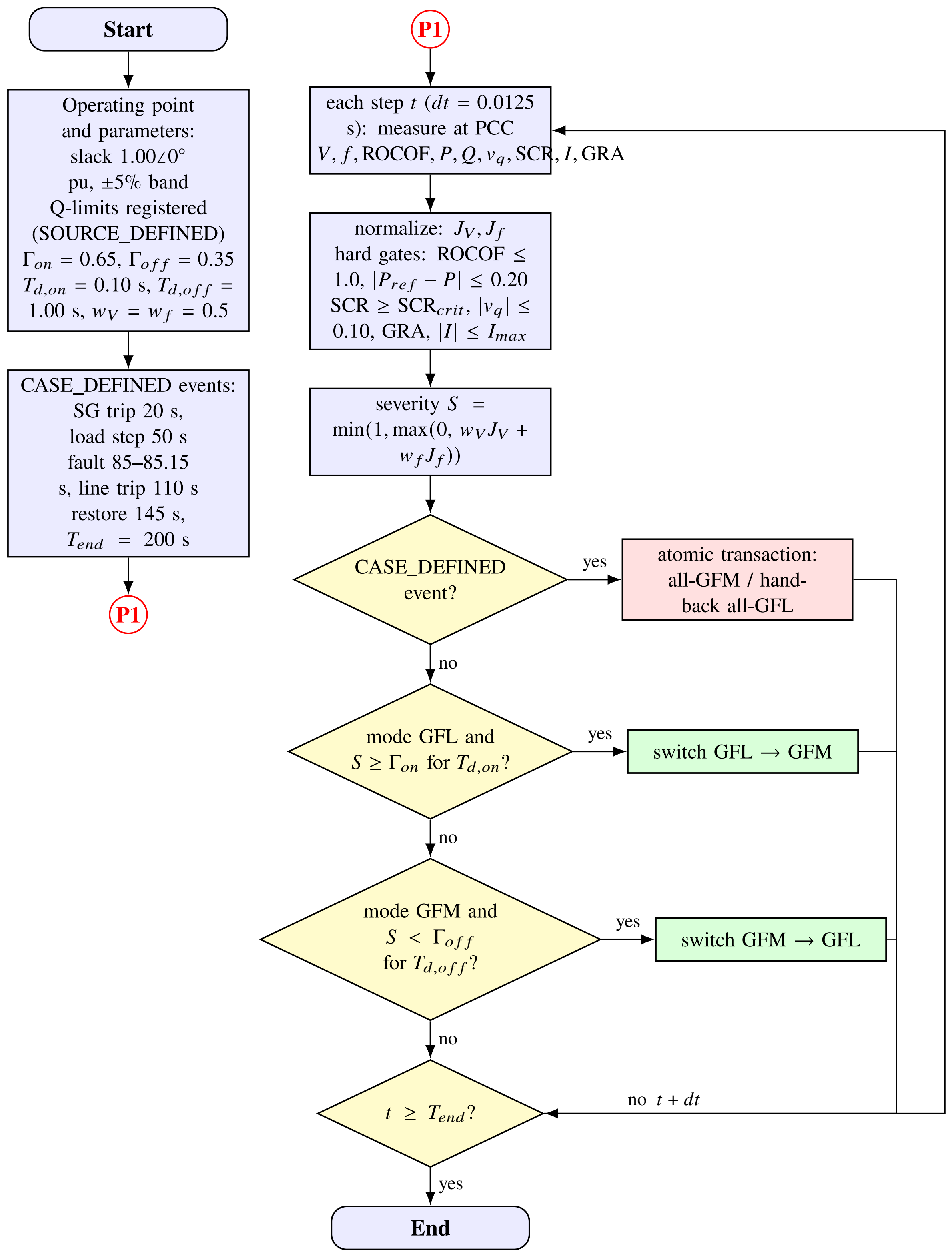

Figure 3: GFL/GFM switching-strategy flowchart. The left column sets the operating point, parameters, and CASE_DEFINED events; the right column is the online loop that measures the PCC signals each step, forms the severity index, and switches mode by threshold and dwell or through the atomic CASE_DEFINED transaction.

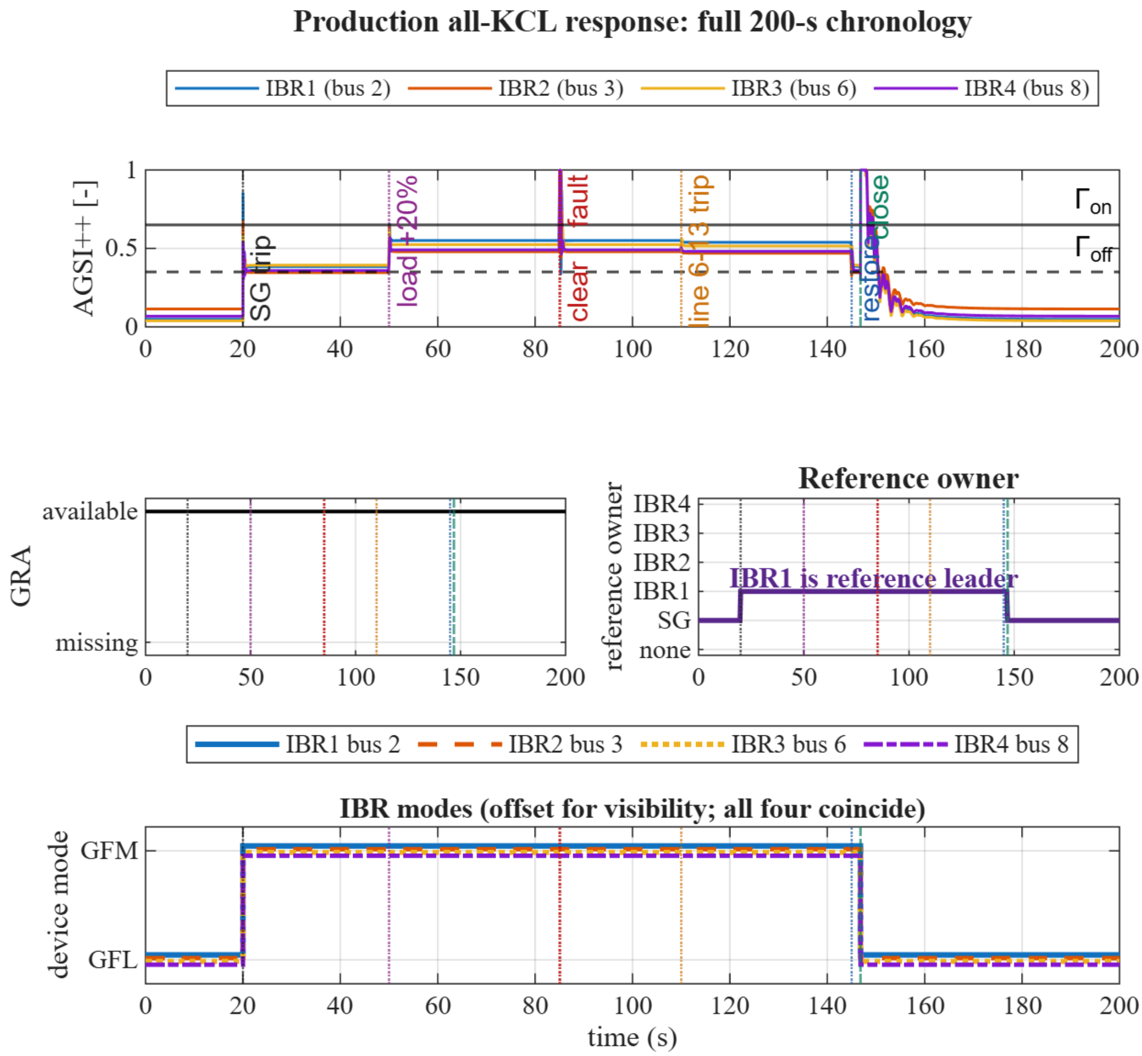

Figure 4: Bounded AGSI++, GRA, reference ownership, and the combined IBR1–IBR4 mode timeline. The owner trace identifies IBR1 at bus 2 as the grid-forming angle/frequency leader while the SG is open. All IBRs are GFL before 20 s, GFM until the accepted handback, and GFL thereafter; the four mode traces are drawn with distinct line styles and a small display-only vertical offset (±0.045 of the 0/1 mode coordinate) because they coincide exactly.

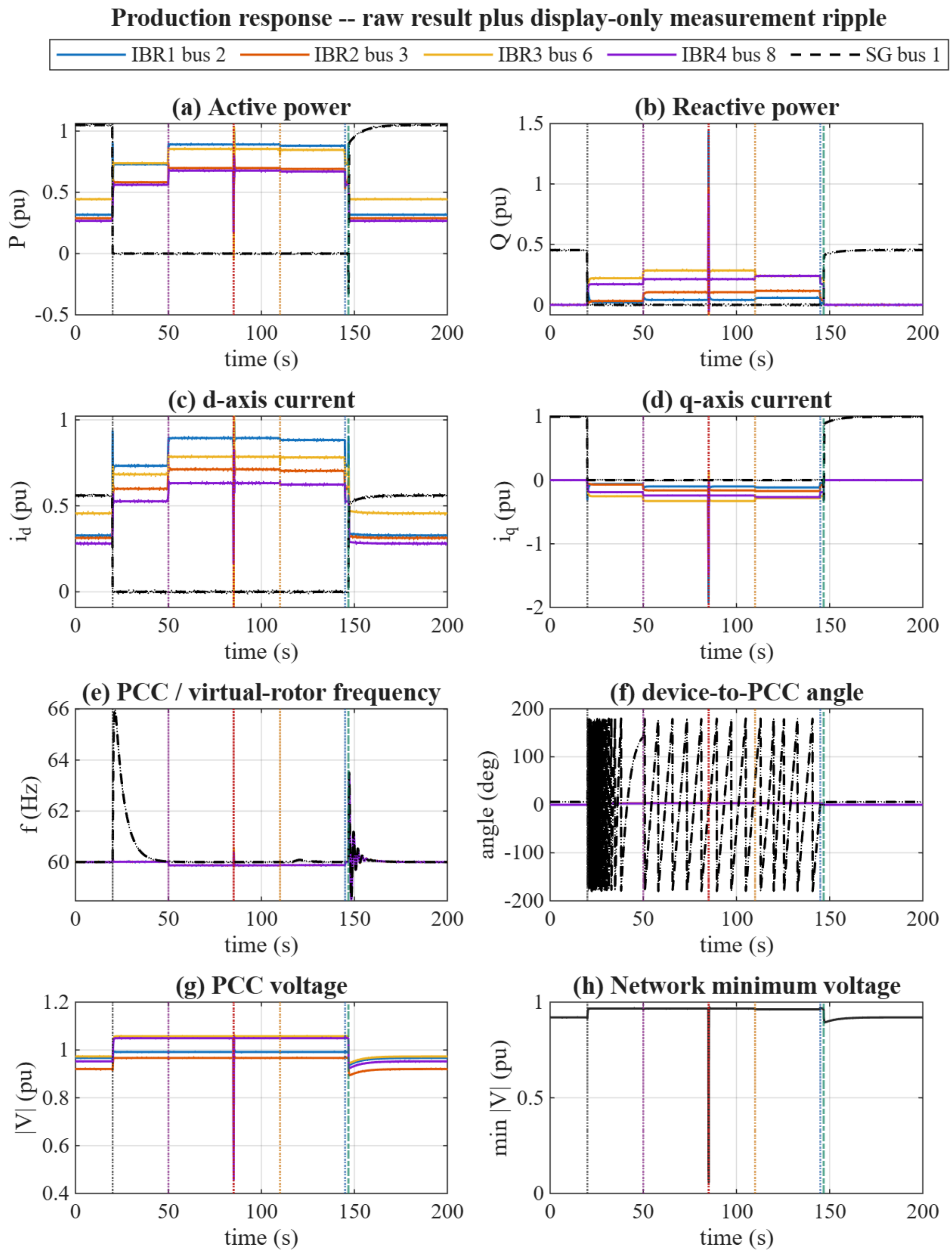

Figure 5: IBR and SG active/reactive power, frame currents, frequency, device-to-PCC angle, PCC voltage, and network minimum voltage. Solid traces are raw production results; dotted band-limited ripple is display-only and does not enter any equation or decision. Remaining vertical event excursions belong to the raw switching transient, not to the ripple.

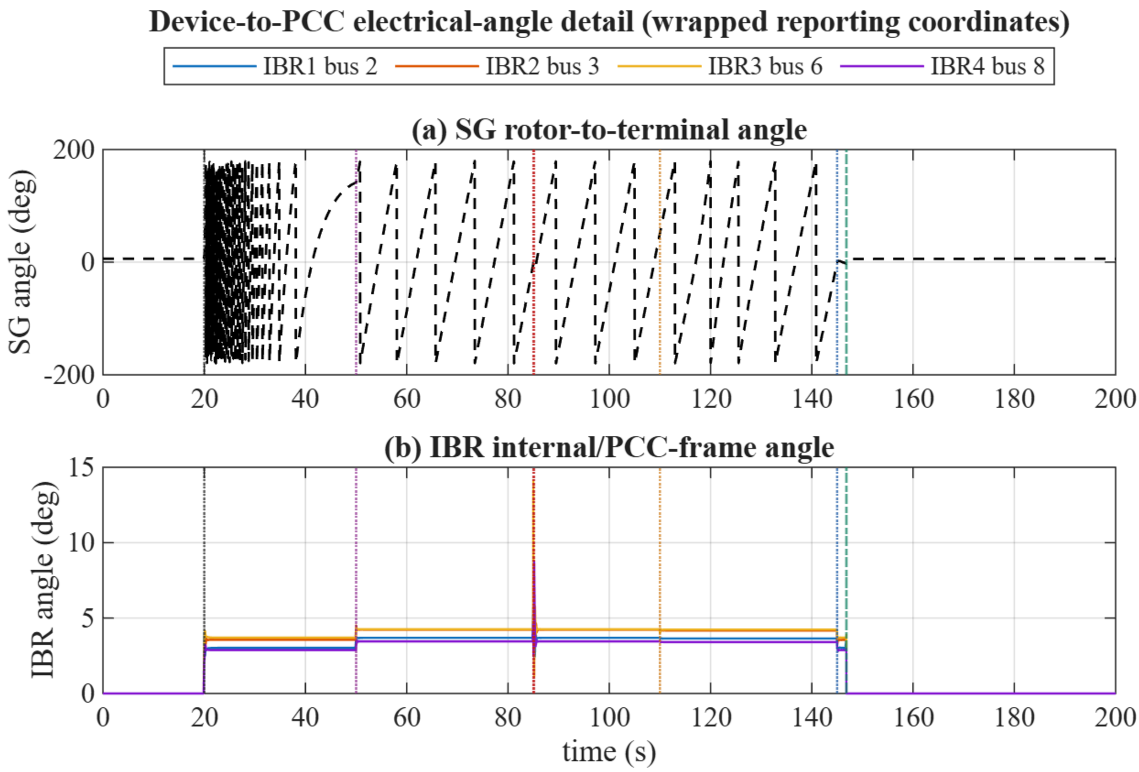

Figure 6: Angle detail. The offline SG rotor slips relative to the energized grid; the plotted coordinate is wrapped to $[-180, 180)$ degrees.

At the accepted close, the synchronism mismatch is $\Delta V = 0.0163$ pu, $\Delta f = 2.75 \times 10^{-6}$ pu, and $\Delta\theta = 3.76°$; all three lie inside the guard limits (0.05 pu, 0.001 pu, 10°). At the 200 s endpoint the all-bus voltage range is 0.9203–1.0000 pu: the SG and IBR regulating buses sit inside the ±5% band, but five remote load buses solve below $V_{\min} = 0.95$ pu. SG electrical power is 1.0497 pu, reactive output is 0.4531 pu, and SG frequency is 60.0002 Hz, i.e. settled from the transient excursion. The trajectory therefore demonstrates a successful declared handback and frequency recovery, while the endpoint reactive profile shows the new operating point has not yet returned every bus to the ±5% band.

6 Evidence boundary and reproduction

The primary run uses $\Delta t = 0.0125$ s. The accepted-close time (146.825 s) is an event-landed output of the synchronism guard, not a scheduled constant, so event-time step-independence is not claimed. The run proves numerical execution of the declared chronology, atomic mode transfer, full KCL, synchronism-gated close, and return to all GFL. It does not validate the diagnostic synchronizer, governor parameters, energy reserve, or a general protection design against field measurements. The dotted display ripple is never used as model evidence; the solid raw traces remain visible.

```
pf_init_paths;
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures(reuse_cache=true);
```

All production solvers are project-owned MATLAB. The targeted numerical, producer and consumer, hybrid event, fail-closed, and external-solver-prohibition suites pass 108/108 tests. The full repository regression was intentionally omitted under the repository risk policy because these targeted gates cover the changed producer, consumers, and failure paths.